

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses S_n for arithmetic sequence with $n = 6$ or $n = 36$	AO1.1a	M1	$S_6 = 3(2a + 5d)$ $= 6a + 15d$ $S_{36} = 18(2a + 35d)$ $= 36a + 630d$
	Finds correct expressions for S_6 and S_{36}	AO1.1b	A1	
	Forms equation in a and d using <i>their</i> $S_{36} = (\text{their } S_6)^2$	AO3.1a	M1	
	Expands quadratic and collects like terms to obtain printed answer Only award for completely correct solution with no errors	AO2.1	R1	$36a + 630d = (6a + 15d)^2$ $36a + 630d = 36a^2 + 90ad + 90ad + 225d^2$ $4a + 70d = 4a^2 + 20ad + 25d^2$
(b)	Uses u_n for arithmetic sequence with $n = 6$	AO1.1b	B1	$a + 5d = 25 \Rightarrow d = \frac{25 - a}{5}$
	Eliminates a or d using <i>their</i> ' $a + 5d = 25$ ' and the printed result in part (a) to obtain a quadratic in one variable	AO1.1a	M1	$4a + 70\left(\frac{25 - a}{5}\right) = 4a^2 + 20a\left(\frac{25 - a}{5}\right) + 25\left(\frac{25 - a}{5}\right)^2$ $4a + 350 - 14a = 4a^2 + 100a - 4a^2 + 625 - 50a + a^2$
	Obtains correct quadratic equation Need not be simplified	AO1.1b	A1	$350 - 10a = 100a + 625 - 50a + a^2$
	Solves their quadratic $a = -5$, $a = -55$ (or $d = 6$, $d = 16$)	AO1.1a	M1	$a^2 + 60a + 275 = 0$
	Deduces min value $a = -55$ NMS $a = -55$ 5/5	AO3.2a	A1	$a = -5$, $a = -55$ (or $d = 6$, $d = 16$) $a = -55$
				Total 9 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
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(a)	Finds a difference between 2 terms	AO3.1a	M1	$3e^p - 5 = 5 - 3e^{-p}$ (*)
	Forms an equation using two differences	AO3.1a	M1	$3e^p - 10 + 3e^{-p} = 0$
	Forms a quadratic equation in e^p	AO1.1a	M1	$3e^{2p} - 10e^p + 3 = 0$
	Obtains a correct quadratic equation	AO1.1b	A1	$e^p = \frac{1}{3}, 3$ $p = \ln \frac{1}{3}, \ln 3$
	Obtains 2 correct solutions for e^p from 'their' quadratic FT only applies if previous mark has been awarded	AO1.1b	A1F	ALT to (*) $2(5 - 3e^{-p}) = 3e^p - 3e^{-p}$ Or $2(3e^{-p} - 5) = 3e^p - 3e^{-p}$
	Obtains final answers in an exact form FT applies if previous mark has been awarded	AO2.2a	A1F	
(b)	Finds a ratio between two consecutive terms (no requirement to use a and r)	AO3.1a	M1	Assume it is possible that $3e^{-q}$, 5 and $3e^q$ are three consecutive terms of a geometric sequence
	Compares two ratios (could be ratios of successive terms, no requirement to use a and r)	AO3.1a	M1	$a = 3e^{-q}, ar = 5, ar^2 = 3e^q$ $\frac{ar}{a} = \frac{5}{3e^{-q}} \Rightarrow r = \frac{5e^q}{3}$ $\frac{ar^2}{ar} = \frac{3e^q}{5} \Rightarrow r = \frac{3e^q}{5}$
	Identifies a contradiction	AO2.1	R1	$\frac{5}{3e^{-q}} = \frac{3e^q}{5} \Rightarrow 25 = 9$
	Draws a conclusion about k	AO2.4	R1	This is a contradiction therefore $3e^{-q}$, 5 and $3e^q$ cannot form three consecutive terms of a geometric sequence.
				Total 10 marks

Q3.

(a) (i) $r = \frac{294}{420} = 0.7$

AG. Accept any valid justification to the given answer

B1

$$(ii) \quad \{S_{\infty} = \} \quad \frac{a}{1-r} = \frac{420}{1-0.7}$$

$$\frac{a}{1-r} \quad \text{used}$$

M1

$$\{S_{\infty} = \} \quad 1400$$

1400 NMS mark as 2 / 2 or 0 / 2

A1

2

$$(iii) \quad nth \text{ term} = 600 \times (0.7)^n$$

If not B2 award B1 for $420 \times (0.7)^{n-1}$ OE

B2

2

$$(b) \quad (i) \quad \{u_n = \} 248 - 8n$$

Accept ACF

B1

1

$$(ii) \quad u_k = 0 \Rightarrow 8k = 248$$

248 - 8k = 0 OE e.g. $240 + (k-1)(-8) = 0$
 ft if no recovery, on c's (b)(i) answer

M1

$$k = 31$$

A1

$$\sum_{n=1}^k u_n = 240 + 232 + \dots + 0 = \frac{k}{2} [240 + 0]$$

For $\frac{k}{2} [240 + 0]$ or for $\frac{k}{2} [c's u_1 + 0]$

OE e.g. $\frac{k}{2} [2 \times c's u_1 + (k-1)(-8)]$

M1

$$\sum_{n=1}^k u_n \quad (= 15.5 \times 240) = 3720$$

3720

A1

4

[10]

Q4.

(a) $\{S_{25} = \frac{25}{2} [2a + (25 - 1)d]$
 $\frac{25}{2} [2a + (25 - 1)d] \quad OE$

M1

$$\frac{25}{2} [2a + 24d] = 3500$$

$$25(2a + 24d) = 7000 \text{ or } \left[\frac{50a + 600d}{2} = 3500 \right]$$

Forming equation and attempt to remove fraction or to expand brackets or better

M1

$$50a + 600d = 7000 \text{ (or better) so } a + 12d = 140$$

CSO AG Be convinced.

A1

3

(b) $5^{\text{th}} \text{ term} = a + 4d$
 $a + (5 - 1)d \text{ used correctly}$

M1

$$a + 12d = 140, a + 4d = 100 \Rightarrow 8d = 40$$

Solving $a + 12d = 140$ simultaneously with either $a + 4d = 100$ or $a + 5d = 100$ as far as eliminating either a or d .

M1

$$\Rightarrow d = 5$$

A1

$$\Rightarrow a = 80$$

A1

4

(c) $33 \left(3500 - \sum_{n=1}^k u_n \right) = 67 \sum_{n=1}^k u_n$

Recognition that $\sum_{n=1}^{25} u_n = 3500$

M1

$$33 \times 3500 = 67 \sum_{n=1}^k u_n + 33 \sum_{n=1}^k u_n$$

Correct rearrangement PI

M1

$$100 \times \sum_{n=1}^k u_n = 33 \times 3500 \Rightarrow \sum_{n=1}^k u_n = 1155$$

A1

3

[10]

Q5.

- (a) (common difference) = 9
9

B1

1

- (b) (100th term) = $23 + (100 - 1)d$
 $23 + (100 - 1)d$ or better seen (or used with $d = 9$
 or with $d = c$'s answer (a))

M1

$$= 914$$

914 NMS mark as B2 or B0

- (c) (Sum of series) = $\frac{280}{2} (23 + 2534)$
Substitution of $n = 280$, $l = 2534$,
 $a = 23$ (or c 's value of a used in (b)),
 $d = 9$ (or c 's answer to (a)) into $\frac{n}{2}(a + l)$

$$\left\{ \text{or } \frac{280}{2} [2 \times 23 + (280 - 1)(9)] \right\}$$

$$PI \text{ or } \frac{n}{2} [2a + (n - 1)d] \quad PI$$

M1

$$= 357\,980$$

357 980 NMS mark as B2 or B0

A1

2

[5]

Q6.

(a) $\{S_{31}\} = \frac{31}{2}[2a + (31-1)d]$

M1

$$31(a + 15d) = 310$$

Forming eqn and eliminating fraction or bracket

m1

$$a + 15d = 310/31; \quad a + 15d = 10$$

AG Completion to printed answer

A1

3

(b) $a + (21-1)d = 2[a + (16-1)d]$

$a + (n-1)d$ used for at least one term

M1

$$\Rightarrow a = -10d; \Rightarrow -10d + 15d = 10$$

Solving $a + 15d = 10$ simultaneously

with an eqn in a and d obtained from

$a + 20d = k[a + 15d]$ with $k = 2$ or with $k = 1/2$

m1

$$d = 2$$

A1

3

(c) $u_1 = a = -20$

ft on c's value for d in $a + 15d = 10$ or in another correct (dep on m1) equation in a and d

The value for a must appear within c's soln for (c)

B1F

$$\sum_{n=1}^k u_n = S_n = \frac{k}{2}[2a + (k-1)d]$$

Condone n for k in M1 and A1F lines

provided n replaced by k at a later stage

M1

$$\frac{k}{2}[-40 + 2k - 2] = 0$$

'= 0' can be implied by later line; ft on c's non-zero values for a and d

A1F

$$k = 21$$

*Condone presence of $k = 0$
SC NMS $k = 21$ and with $d = 2$ found
 earlier award B2. If $k = 21$ but never see
 $d = 2$, award 0/4*

A1

4

[10]

Q7.

(a) (i) $\{S_{\infty}\} = \frac{a}{1-r} = \frac{10}{1-r}$
 $\frac{a}{1-r}$ used

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M1

$$\frac{10}{1-r} = 50 \text{ so } 1-r = \frac{10}{50} \Rightarrow r = \frac{4}{5}$$

AG Condone verification with the correct final statement but be convinced.

A1

2

(ii) $2^{\text{nd}} \text{ term} = ar$

ar stated or used for the 2^{nd} term. PI by ans'8'

M1

$$= 8$$

A1

2

- (b) (i) 4th term = $a + 3d$; 8th term = $a + 7d$
Uses $a + (n - 1)d$ correctly at least once

M1

$$a + 3d = 10, \quad a + 7d = 8$$

Both eqns. correct ft on c's (a)(ii) OE eg
 $8 = 10 + 4d$

A1F

$$\Rightarrow 4d = -2 \Rightarrow d = -0.5$$

OE fraction.

A1

3

- (ii) $a + 3(-0.5) = 10$
An appreciation that a is required in
***(b)(ii)** and a valid method to find a*
anywhere or PI if $a = 11.5$ seen/used

M1

$$\Rightarrow a = 11.5$$

A1F

Ft on c's non-zero value for d
ie using $a = 10 - 3d$ or $a = c's\ 8 - 7d$.
[c's 8 is candidate's answer to (a)(ii)]

$$\sum_{n=1}^{40} u_n = S_{40} = \frac{40}{2} [2a + (40 - 1)d]$$

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$$\frac{40}{2} [2a + (40 - 1)d] \quad \text{OE}$$

M1

$$= 70$$

A1

4

[11]

Q8.

(a) $\{S_{40} = \frac{40}{2} [2a + (40 - 1)d]$

M1

$$20(2a + 39d) = 1250$$

A1

$$\{25^{\text{th}} \text{ term} \Rightarrow a + (25 - 1)d$$

M1

$$a + 24d = 38$$

A1

Dep on both previous two Ms. Solving two equations in a and d simultaneously

m1

$$18d = 27 \Rightarrow d = 1.5$$

AG Be convinced

SC Using the given answer for d : mark out of a maximum of 4/6 as

M1A1M1A1 {conclusion also needed in last A mark} (m0A0)

A1cso

6

$$(b) \quad a = 38 - 24 \times 1.5 \\ = 2$$

PI if using $a = 2$ in (b)

If using eg $a = 38$ award this M mark at

stage: no. of terms $\frac{100 - 38}{1.5} + 1 + 24$

M1

$$a + (n - 1) 1.5 < 100$$

M1

$$n < \frac{100 - a}{1.5} + 1$$

$$n < 66.333 \dots$$

$\Rightarrow$ number of terms < 100 is 66

NMS mark as B3 for 66 else B0

A1

3

[9]

Q9.

(a) $(d) = 7$

B1

1

(b) $(101^{\text{st}} \text{ term}) = a + (101 - 1) d$

M1

..... = $51 + 100(7) = 751$

*Ft on c's answer for d. NMS/rep. addn.,
give both marks for '751'.*

SC

if M0, award B1 For $7n + 44$ OE

A1F

2

(c) $S_n = \frac{100}{2} [751 + 1444]$ **or**

$S_n = \frac{100}{2} [2 \times 751 + (100 - 1)7]$

*Formula for $\{S_n\}$ with [any 3 of
 $a = c$'s 751 (condoning '751' $\pm d$)
or $d = c$'s 7 **or** $n = 100$ **or**
 $l = 1444$ substituted] or $[S_{200} - S_k$ with
 $k = 100$, (condoning $k = 99$ or 101)
stated/used with correct ft substitution in
 S_{200} or $S_k]$*

M1

= 109 750

A1

2

[5]

Q10.

(a) $\{S_{29}\} = \frac{29}{2} [2a + 28d]$

Formula for S_n with $n = 29$ substituted and with a and d

M1

$29(a + 14d) = 1102$

Equation formed then some manipulation

m1

$$a + 14d = \frac{1102}{29} \Rightarrow a + 14d = 38$$

CSO AG

A1

3

(b) $u_2 = a + d \quad u_7 = a + 6d$
Either expression correct

B1

$$u_2 + u_7 = 13 \Rightarrow 2a + 7d = 13$$

Forming equation using u_2 & u_7 both in form $a + kd$

M1

e.g. $21d = 63; 3a = -12$
*Solving $a + 14d = 38$ with candidate's ' $2a + 7d = 13$ '
 to at least stage of elimination of either a or d*

m1

$a = -4 \quad d = 3$
Both correct

A1

4

[7]

Q11.

(a) (Tenth term) $= a + (10 - 1)d$

M1

..... $= 1 + 9(6) = 55$
NMS or rep. addition. B2 CAO
SC if M0 award B1 for $6n - 5$ OE

A1

2

(b) (i) $S_n = \frac{n}{2} [2 + (n - 1)6]$
Formula for $\{S_n\}$ with either $a = 1$ or $d = 6$ substituted

M1

$$\frac{n}{2} [2 + 6n - 6] = 7400$$

Equation formed with some expansion of brackets

A1

$$3n^2 - 2n = 7400 \Rightarrow 3n^2 - 2n - 7400 = 0$$

CSO AG

A1

3

(ii) $(3n + 148)(n - 50) = 0$

Formula/factorisation OE

M1

$$\Rightarrow n = 50$$

NMS single ans. 50.. B2 CAO

NMS 50 and -49.3(3...) B1 CAO

A1

2

[7]

Q12.

(a) (i) $a + 19d = 181$;

$$a + 4d = 46$$

$$a + (n-1)d \text{ used; PI}$$

M1

$$\Rightarrow 15d = 181 - 46$$

A1

$$\Rightarrow d = 9$$

AG (be convinced)

A1

3

(ii) $a = 10$

B1

1

$$(b) \quad S_{20} = \frac{20}{2} [2a + (20-1)d]$$

M1

$\dots = 1910$

A1

2

$$(c) \quad \sum_{n=1}^{50} \mathcal{L}_n - \sum_{n=1}^{20} \mathcal{L}_n$$

M1

$$.... = 11525 - "1910" = 9615$$

ft on 11525 – c's S_{20}

A1ft

2

[8]

Q13.

(a) $u_1 = 87; u_2 = 84$

ft on $u_2 = u_1 - 3$
SC B1 for 90, 87

B1;B1ft

2

(b) Common difference (d) is -3

B1

1

(c) $\sum_{n=1}^k 2d_n$ = sum of AP

M1

$$\dots\dots\dots = \frac{k}{2} [174 + (k-1)(-3)]$$

*OE ft on u_1 and use of $d = 3$
(For M1A1 ft condone n in place of k)*

A1ft

$$0 = \frac{k}{2} [177 - 3k] \Rightarrow 177 = 3k$$

$$\Rightarrow k = 59$$

*Just the **single** value 59*

A1

3

$$\text{ALT1} = \sum_{n=1}^k 90 - \sum_{n=1}^k 3n = 90k - 3 \left[\frac{k}{2} (k+1) \right]$$

M1 split and either $90k$ or $\left[\frac{k}{2} (k+1) \right]$

M1; A1

$$0 = 90k - 1.5k(k+1) \Rightarrow k = 59$$

(For 1st two marks condone n in place of k)

A1

[6]



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